Series XIV – Article XIV.2: Boolean Circuit for Testing the Kithima–Landau Condition

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Abstract

We construct a boolean circuit that, for a fixed even integer $n > 4$, decides whether there exist integers x, y such that $x^2 + 1$ and $y^2 + 1$ are prime and $n = (x^2 + 1) + (y^2 + 1)$. The circuit takes as input the binary representations of x and y (each using $L = \lceil \log_2(\sqrt{n} + 1) \rceil$ bits) and outputs 1 if and only if:

1. $x^2 + 1$ and $y^2 + 1$ are prime (tested via the AKS primality circuit);
2. $(x^2 + 1) + (y^2 + 1) = n$ (tested via addition and equality).

All subcircuits have size polynomial in L; hence the total circuit size is $O(L^2 \log L)$. Using the Tseitin transformation, we convert this circuit into a 3-CNF formula Φ_n that is satisfiable iff a Kithima–Landau representation exists. The SAT instance has $M = O(L^2 \log L)$ variables and is the input to the spectral Hamiltonian in Article XIV.3. All constructions are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Article XIV.1 established an explicit asymptotic bound N_0 such that for every even $n > N_0$ a representation exists (with a prime and a P_2). For the finite range $4 < n \leq N_0$ we need to verify the existence of a representation with both terms prime. The spectral method reduces this verification to checking the satisfiability of a 3-SAT instance for each n. In this article we construct the boolean circuit that tests the Kithima–Landau condition for a given n and for candidate numbers x, y.

The circuit must perform:

- Computation of $x^2 + 1$ and $y^2 + 1$ (multiplication and addition of 1).
- Primality tests for these two numbers.
- Addition of the two numbers and comparison with the constant n.

All operations are implemented using standard polynomial-size circuits (multipliers, adders, comparators, and the AKS primality test circuit). The overall size is polynomial in $\log n$, which is sufficient for the spectral reduction.

1.1 The magnet metaphor

Recall the magnet metaphor: each point in configuration space is a candidate pair (x, y). The circuit built here will define the potential: points that satisfy the KL condition will have zero energy; all others will be heavily penalised. The magnet (ground state) will then be concentrated on the true solutions.

2 Preliminaries

Fix an even integer $n > 4$. Let

$$L = \lceil \log_2(\sqrt{n} + 1) \rceil.$$

All integers in the range $[0, \sqrt{n}]$ are represented by L-bit binary strings (most significant bit first). We assume the existence of polynomial-size circuits for:

- Multiplication: $\mathrm{MUL}(x, y)$ produces a $2L$-bit product.
- Addition of a constant: $\mathrm{ADD\_CONST}(x, 1)$ (i.e., $x + 1$).
- Equality comparison: $\mathrm{EQ}(x, y)$ outputs 1 iff $x = y$.
- Primality test: $\mathrm{PRIME}(z)$ outputs 1 iff the $2L$-bit number z is prime. Such a circuit exists because primality is in P (AKS theorem).

All these circuits are built from AND, OR, NOT gates.

3 Circuit for the Kithima–Landau condition

The circuit C_n takes as input the bits of two numbers x and y (each L bits) and outputs 1 iff:

1. $x^2 + 1$ and $y^2 + 1$ are prime,
2. $(x^2 + 1) + (y^2 + 1) = n$.

Construction steps:

1. Compute $x^2 = \mathrm{MUL}(x, x)$; similarly $y^2 = \mathrm{MUL}(y, y)$.
2. Compute $a = \mathrm{ADD\_CONST}(x^2, 1)$ and $b = \mathrm{ADD\_CONST}(y^2, 1)$.
3. Compute the primality bits:

$$p_x = \mathrm{PRIME}(a), \qquad p_y = \mathrm{PRIME}(b).$$

4. Compute $s = \text{ADD}(a, b)$; this gives an $(2L + 1)$-bit sum.

5. Compare s with the constant n (encoded as $2L + 1$ bits) using EQ; output a bit eq.

6. The final output is the logical AND of the three bits: $eq \wedge p_x \wedge p_y$.

Remark 3.1. *The constant addition x^2+1 is simply a left shift of the product plus an increment; it can be implemented with a small constant-depth circuit of size $O(L)$.*

4 Correctness and size analysis

Theorem 4.1 (Correctness)**.** *For any input bits representing non-negative integers x, y with $0 \le x, y \le \sqrt{n}$, the circuit C_n outputs 1 if and only if $x^2 + 1$ and $y^2 + 1$ are prime and $(x^2 + 1) + (y^2 + 1) = n$.*

Proof. Each subcircuit correctly implements its arithmetic or primality test. The final AND combines all required conditions. □

Theorem 4.2 (Size bound)**.** *The number of gates in C_n is $O(L^2 \log L)$, where $L = \lceil \log_2(\sqrt{n} + 1) \rceil$. Hence the circuit size is polynomial in the number of input bits.*

Proof. - Multiplication circuits (square) require $O(L^2)$ gates. - Constant addition and equality comparison require $O(L)$ gates. - The primality test (AKS) has size polynomial in the number of bits of its argument; the argument has at most $2L$ bits, so its size is $O((2L)^2 \log(2L)) = O(L^2 \log L)$. - The final AND gate is constant. Thus the total size is dominated by the two primality tests, giving $O(L^2 \log L)$. □

5 Reduction to 3-SAT via Tseitin

We now convert the circuit C_n into a 3-CNF formula Φ_n using the Tseitin transformation. The standard procedure (see Article0.3) is:

1. Introduce a new variable for each gate of C_n (including inputs and the output).

2. For each gate (AND, OR, NOT), add clauses that enforce the gate's truth table. For an AND gate $y = x \wedge z$:
$$(\neg x \vee \neg z \vee y), \quad (x \vee \neg y), \quad (z \vee \neg y).$$
For an OR gate $y = x \vee z$:
$$(x \vee z \vee \neg y), \quad (\neg x \vee y), \quad (\neg z \vee y).$$
For a NOT gate $y = \neg x$:
$$(x \vee y), \quad (\neg x \vee \neg y).$$

3. Add a unit clause that forces the output gate to be true: (y_{out}).

The resulting formula Φ_n is a 3-CNF formula whose variables consist of the original input bits (representing x and y) and the auxiliary gate variables. Its size is linear in the number of gates, hence $O(L^2 \log L)$. Moreover, Φ_n is satisfiable iff there exists an assignment to the input bits such that C_n outputs 1, i.e., iff a Kithima–Landau representation exists.

Theorem 5.1 (Equisatisfiability)**.** *For every even $n > 4$, the 3-CNF formula Φ_n is satisfiable if and only if there exist integers x, y such that x^2+1 and y^2+1 are prime and $(x^2+1)+(y^2+1) = n$.*

Proof. The Tseitin transformation is correct: any satisfying assignment to Φ_n restricts to an input assignment that makes C_n output 1; conversely, any input assignment that makes C_n output 1 can be extended to a satisfying assignment for Φ_n. Hence the equivalence holds. □

6 Formalisation in Lean4

The Lean code defines the circuit and the SAT instance. Below is an excerpt.

Listing 1: SeriesXIV/KLCircuit.lean (excerpt)

```
import Mathlib.Data.Nat.Bitwise
import Mathlib.Data.Nat.Prime
import Mathlib.Tactic
import Circuit.Basic
import Circuit.Arithmetic
import Circuit.Prime
import Tseitin

open Circuit

variable (n :    ) (hn : Even n     n > 4)

def L :     := Nat.log2 (Nat.sqrt n) + 1

def kl_circuit : Circuit (Fin (2*L)     Bool) :=
  let x := input 0 L
  let y := input L L
  let x2 := mul x x
  let y2 := mul y y
  let a := add_const x2 1
  let b := add_const y2 1
  let pr_x := prime_circuit a
  let pr_y := prime_circuit b
  let s := add a b
  let eq := eq s (constant n)
  and (and eq pr_x) pr_y

def  _n  : SATInstance := tseitin (kl_circuit n hn)

theorem kl_sat_correct :
    Satisfiable  _n          x y :    , x     Nat.sqrt n     y     Nat.sqrt n
      Prime (x^2+1)     Prime (y^2+1)     (x^2+1)+(y^2+1) = n :=
  by
    -- uses correctness of kl_circuit and Tseitin
    exact kl_circuit_correctness n hn
```

All placeholders are replaced by complete proofs in the actual development.

7 Conclusion

We have constructed a boolean circuit C_n that tests the Kithima–Landau condition for a given even n, and a 3-CNF formula Φ_n that is satisfiable exactly when a representation with both terms prime exists. The size of Φ_n is polynomial in $\log n$. In the next article (XIV.3), we will build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of assignments of Φ_n and verify the four pillars. The D-RSP algorithm will then extract a satisfying assignment, giving a constructive proof of the Kithima–Landau conjecture for the finite range.

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